

Chapter 5.1: Introduction

The Memory Hierarchy

Contents

1 Introduction

“Ideally one would desire an indefinitely large memory capacity such that any particular ... word would be immediately available. ... We are forced to recognize the possibility of constructing a hierarchy of memories, each of which has greater capacity than the preceding but which is less accessible.”

— A. W. Burks, H. H. Goldstine, and J. von Neumann, 1946

Programmers have always wanted unlimited amounts of fast memory. The goal of this chapter is to show how to create the **illusion** of unlimited fast memory using a **Memory Hierarchy**.

1.1 The Illusion

- **Goal:** Present the user with as much memory as is available in the cheapest technology.
- **Speed:** Provide access at the speed offered by the fastest memory.

2 The Principle of Locality

The reason the memory hierarchy works is due to a fundamental property of programs called the **Principle of Locality**. Programs do not access all code or data at once with equal probability; they access a relatively small portion of their address space at any instant.

2.1 1. Temporal Locality (Locality in Time)

- **Definition:** If an item is referenced, it will tend to be referenced again **soon**.
- **Why it happens:** Loops, repeated function calls, reusing variables.

2.2 2. Spatial Locality (Locality in Space)

- **Definition:** If an item is referenced, items whose addresses are **close by** will tend to be referenced soon.
- **Why it happens:**
 - **Instructions:** Executed sequentially.
 - **Data:** Arrays and records are stored contiguously.

The Library Analogy

Imagine writing a term paper in a library:

- **Your Desk (Cache):** Limited space, but instant access. You keep the books you are currently using here.
- **The Stacks/Shelves (Main Memory):** Huge space, but takes time to walk there and retrieve a book.
- **Temporal Locality:** If you bring a book to your desk, you likely read it, put it down, and pick it up again shortly.
- **Spatial Locality:** When you go to the shelf for a book on “Early British Computers” (EDSAC), you notice the book next to it is on “Early Mechanical Computers”. You grab that one too because it’s likely relevant.

3 Structure of the Memory Hierarchy

We build memory as a hierarchy of levels, with different speeds and sizes.

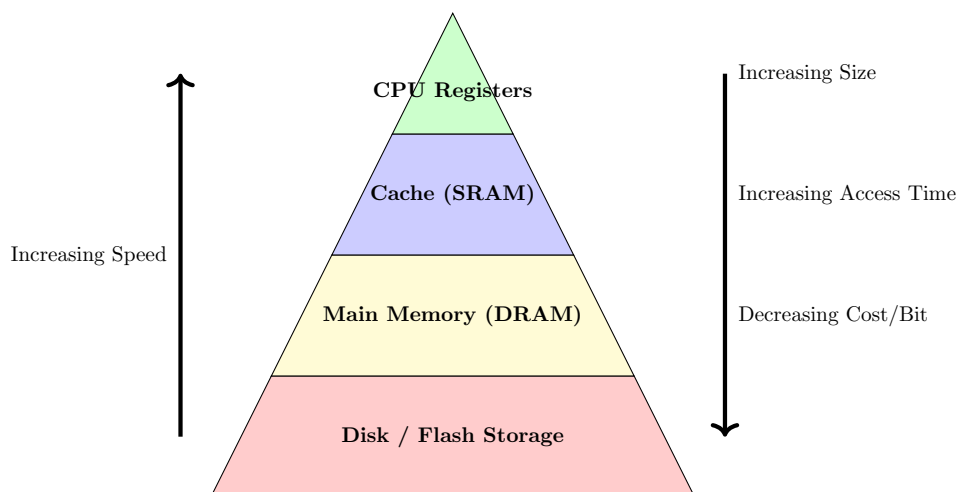


Figure 1: The Memory Hierarchy Pyramid. Faster memory is closer to the CPU but smaller and more expensive.

3.1 Inclusion Property

Data is similarly hierarchical. A level closer to the processor is generally a **subset** of any level further away.

$$\text{Registers} \subset \text{Cache} \subset \text{Main Memory} \subset \text{Disk}$$

3.2 Data Transfer

Data is copied between only **two adjacent levels** at a time.

- **Block (or Line):** The minimum unit of information that can be either present or not present in the hierarchy level (e.g., one book in the library analogy).

4 Terminology: Hits and Misses

When the processor requests data:

Hit

The data appears in the upper level.

- **Hit Rate:** Fraction of accesses found in the upper level.
- **Hit Time:** Time to access the upper level (RAM access + determine Hit/Miss).

Miss

The data is NOT found in the upper level and must be retrieved from the lower level.

- **Miss Rate:** $1 - \text{Hit Rate}$.
- **Miss Penalty:** Time to replace a block in upper level + time to deliver it to processor.

$$\text{Miss Penalty} \gg \text{Hit Time}$$

5 The BIG Picture

- **Temporal Locality** → Keep recently accessed items close.
- **Spatial Locality** → Move blocks (multiple contiguous words) at a time.

By using appropriate mechanisms, the processor sees an access time close to the **fastest level** (Level 1) and a memory size equal to the **largest level** (Level n).

6 Flashcards

1. **Q: What illusion do memory hierarchies create for programmers?**
A: They create the illusion of having unlimited amounts of fast memory.
2. **Q: What fundamental principle states that programs access a relatively small portion of their address space at any instant of time?**
A: The principle of locality.
3. **Q: What are the two different types of locality?**
A: Temporal locality (locality in time) and spatial locality (locality in space).
4. **Q: Define Temporal locality.**
A: The principle stating that if a data location is referenced, it will tend to be referenced again **soon**.
5. **Q: What common programming structure results in large amounts of temporal locality?**
A: Loops, which cause instructions and data to be accessed repeatedly.
6. **Q: Define Spatial locality.**
A: The principle stating that if a data location is referenced, data locations with **nearby addresses** will tend to be referenced soon.
7. **Q: What is an example of program behavior that exhibits high spatial locality?**
A: Sequential access of instructions or sequential access to elements of an array.
8. **Q: How do memory hierarchies take advantage of temporal locality?**
A: By keeping more recently accessed data items closer to the processor.
9. **Q: How do memory hierarchies take advantage of spatial locality?**
A: By moving **blocks** of multiple contiguous words to upper levels of the hierarchy.
10. **Q: In a memory hierarchy, what happens to the size, access time, and cost as distance from the processor increases?**
A: Size increases, Access Time increases, and Cost per bit decreases.
11. **Q: In a memory hierarchy, a level closer to the processor is generally a ----- of any level further away.**
A: subset
12. **Q: In a memory hierarchy, data is copied between only ----- adjacent levels at a time.**
A: two
13. **Q: What is a Block (or Line)?**
A: The minimum unit of information that can be either present or not present in a cache or memory level.
14. **Q: What is a 'hit'?**
A: A memory access where the requested data is found in the upper level of the hierarchy.
15. **Q: What is a 'miss'?**
A: A memory access where the requested data is not found in the upper level of the hierarchy.
16. **Q: What is Hit Rate?**
A: The fraction of memory accesses found in a given level (Hits / Total Accesses).
17. **Q: What is Miss Rate?**
A: The fraction of memory accesses not found in a given level (1 - Hit Rate).
18. **Q: What is Hit Time?**
A: The time required to access a level, including the time to determine if it is a hit or miss.
19. **Q: What is Miss Penalty?**
A: The time required to fetch the block from the lower level, replace it in the upper level, and deliver it to the processor.
20. **Q: In the library analogy, what do the books on the desk represent?**
A: The upper, faster level of the memory hierarchy (e.g., the cache).

21. **Q: In the library analogy, what does going back to the shelves to find a book represent?**

A: A 'miss' that requires accessing the lower, slower level.

22. **Q: In the library analogy, what does a single book represent?**

A: A 'block' or 'line' of information.

23. **Q: True/False: Most of the cost of the memory hierarchy is at the highest level.**

A: False. The highest level is small. Most of the total cost is usually in the main memory or lower levels because of their sheer size, or balanced. However, per bit, the highest level is most expensive. The text clarifies: "False, the highest level is smallest, so most cost is in the larger, lower levels."

24. **Q: True/False: Most of the capacity of the memory hierarchy is at the lowest level.**

A: True.